

Set 5: Website Analytics

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COURSE CODE- DSA 0603

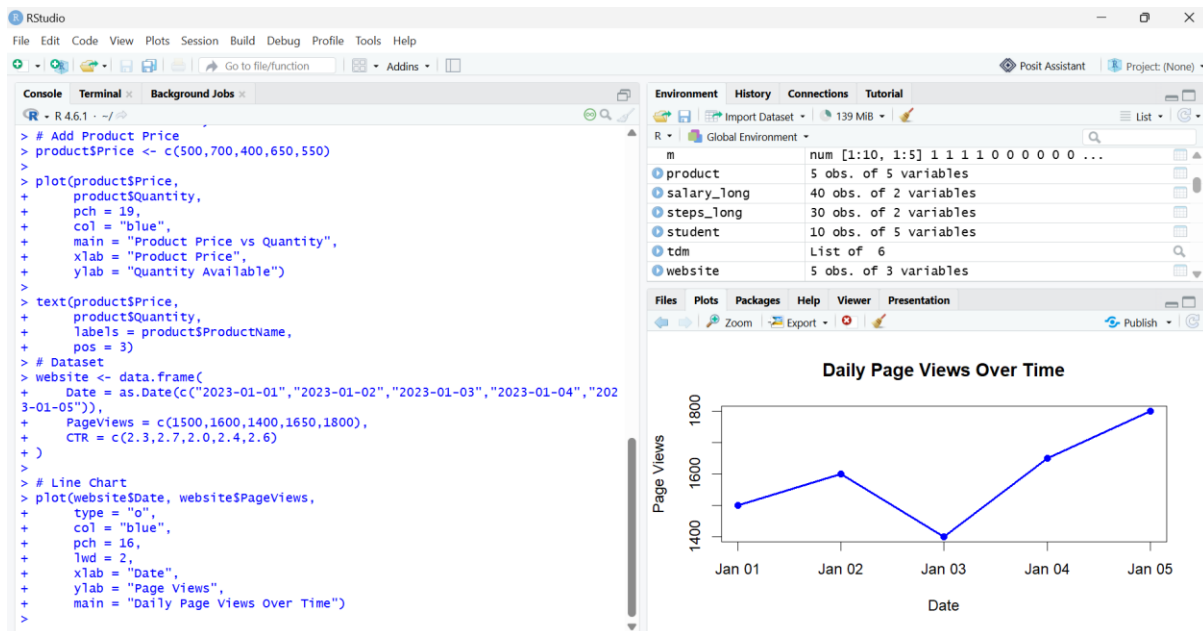
COURSE NAME- DATA HANDLING AND VISUALISATIONS

Question 1: Line Chart (Daily Page Views)

Code:

```
website <- data.frame(  
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04", "2023-01-05")),  
  PageViews = c(1500, 1600, 1400, 1650, 1800),  
  CTR = c(2.3, 2.7, 2.0, 2.4, 2.6)  
)  
plot(website$Date, website$PageViews,  
  type = "o",  
  col = "blue",  
  pch = 16,  
  lwd = 2,  
  xlab = "Date",  
  ylab = "Page Views",  
  main = "Daily Page Views Over Time")
```

Output:

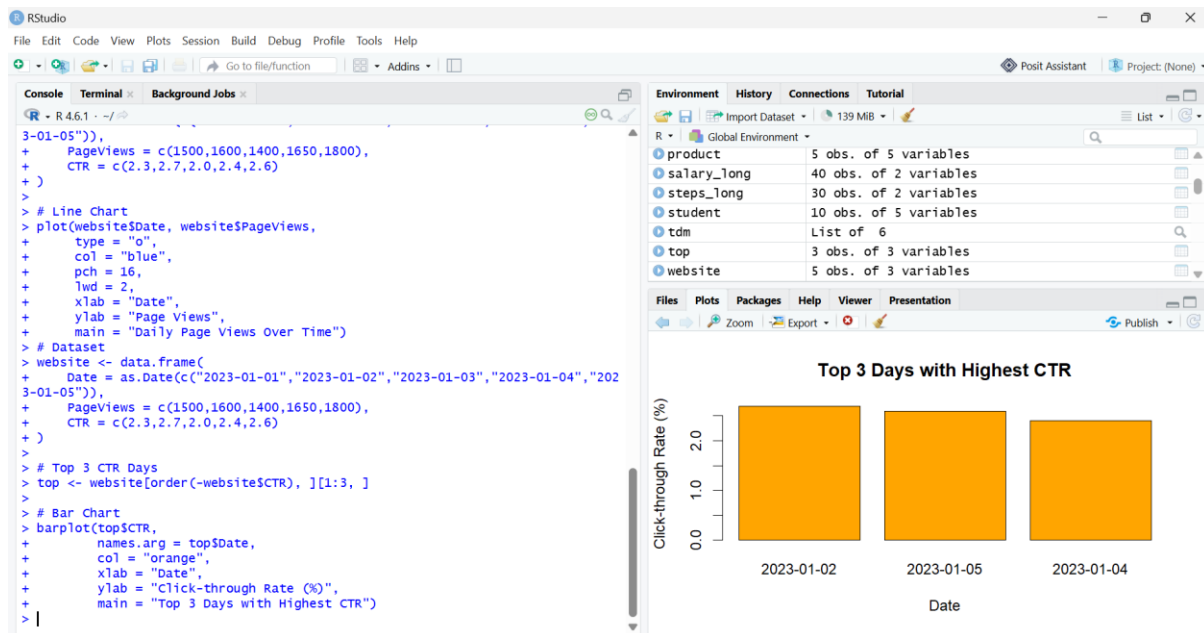


Question 2: Bar Chart (Top 3 Days with Highest CTR)

Code:

```
website <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04", "2023-01-05")),
  PageViews = c(1500,1600,1400,1650,1800),
  CTR = c(2.3,2.7,2.0,2.4,2.6))
top <- website[order(-website$CTR), ][1:3, ]
barplot(top$CTR,
  names.arg = top$Date,
  col = "orange",
  xlab = "Date",
  ylab = "Click-through Rate (%)",
  main = "Top 3 Days with Highest CTR")
```

Output:



Question 3: Stacked Area Chart (Likes, Shares, Comments)

Code:

```
date <- as.Date(c("2023-01-01","2023-01-02","2023-01-03","2023-01-04","2023-01-05"))
```

```
likes <- c(300,350,280,370,400)
```

```
shares <- c(120,150,100,140,160)
```

```
comments <- c(80,90,70,85,95)
```

```
library(plotrix)
```

```
stackpoly(date,
```

```
  cbind(likes, shares, comments),
```

```
  col = c("skyblue","lightgreen","pink"),
```

```
  xlab = "Date",
```

```
  ylab = "User Interactions",
```

```
  main = "Stacked Area Chart")
```

```
legend("topleft",
```

```
  legend = c("Likes","Shares","Comments"),
```

```
  fill = c("skyblue","lightgreen","pink"))
```

Output:

